



PAMANTASAN NG LUNGSOD NG SAN PABLO
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NUMERICAL METHODS

CPE 312

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Lecturer



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NEWTON-RAPHSON METHOD



Learning Objectives

- *Define Newton-Raphson Method
- *Determine and perform the algorithm of Newton-Raphson Method
- *Solve functions using Newton-Raphson Method



- **Newton Raphson Method or Newton Method** is a powerful technique for solving equations numerically.
- It is most commonly used for approximation of the roots of the real-valued functions.



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- Newton Raphson Method was developed by Isaac Newton and Joseph Raphson, hence the name Newton Raphson Method.
- This method is not efficient in calculating the roots of the polynomials or equations with higher degrees but in the case of small-degree equations, this method yields very quick results.



Newton-Raphson Method Algorithm

1. Find points a and b such that $a < b$ and $f(a) \cdot f(b) < 0$.
2. Take the interval $[a, b]$ and find the next value using $x_0 = \frac{a+b}{2}$.



3. Find $f(x_0)$ and $f'(x_0)$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

4. If $f(x_0) \approx 0$, then x_1 is the approximate root.

Stopping criterion (0.000...)



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Example-1

Find a root of $f(x) = x^3 - x - 1$ using Newton Raphson method.



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Step 1.

$$f(x) = x^3 - x - 1$$



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$$f(2) = 2^3 - 2 - 1 = 5$$



Step 1.

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Step 1.

$$f(x) = x^3 - x - 1$$

$$f(0) = 0^3 - 0 - 1 = -1$$

$$f(1) = 1^3 - 1 - 1 = -1$$

$$f(2) = 2^3 - 2 - 1 = 5$$

Initial Interval **[1, 2]**



Step 2.

$$X_0 = \frac{a+b}{2}$$

$$X_0 = \frac{1+2}{2}$$

$$X_0 = \frac{3}{2}$$

$$X_0 = 1.5$$



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<i>Iterations</i>	x_0	$f(x_0)$	$f'(x_0)$	x_1
1	1.5			



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$$f(x) = x^3 - x - 1$$

$$f(1.5) = (1.5)^3 - 1.5 - 1$$



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$$f(x) = x^3 - x - 1$$

$$f(1.5) = (1.5)^3 - 1.5 - 1$$

$$\mathbf{f(1.5) = 0.875}$$



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<i>Iterations</i>	x_0	$f(x_0)$	$f'(x_0)$	x_1
1	1.5	0.875		



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$$f(x) = x^3 - x - 1$$

$$f'(x) =$$



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$$f(x) = x^3 - x - 1$$

$$f'(x) = 3x^2 - 1$$



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$$f(x) = x^3 - x - 1$$

$$f'(x) = 3x^2 - 1$$

$$f'(1.5) = 3(1.5)^2 - 1$$



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$$f(x) = x^3 - x - 1$$

$$f'(x) = 3x^2 - 1$$

$$f'(1.5) = 3(1.5)^2 - 1$$

$$f'(x) = 5.75$$



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<i>Iterations</i>	x_0	$f(x_0)$	$f'(x_0)$	x_1
1	1.5	0.875	5.75	



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Step 3.

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$



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Step 3.

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 1.5 - \frac{0.875}{5.75}$$

$$\mathbf{x_1 = 1.34783}$$



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<i>Iterations</i>	x_0	$f(x_0)$	$f'(x_0)$	x_1
1	1.5	0.875	5.75	1.34783



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<i>Iterations</i>	x_0	$f(x_0)$	$f'(x_0)$	x_1
1	1.5	0.875	5.75	1.34783
2	1.34783			



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<i>Iterations</i>	x_0	$f(x_0)$	$f'(x_0)$	x_1
1	1.5	0.875	5.75	1.34783
2	1.34783	0.10070		



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<i>Iterations</i>	x_0	$f(x_0)$	$f'(x_0)$	x_1
1	1.5	0.875	5.75	1.34783
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1	1.5	0.875	5.75	1.34783
2	1.34783	0.10070	4.44994	1.32520



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3	1.32520			



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4	1.32472	0.00001		



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The approximate root of
 $f(x) = x^3 - x - 1$ is 1.32472.



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ACTIVITY # 7



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1. Find the approximate root of $f(x) = \ln(x) - 2$ using Newton-Raphson Method.
2. Use the Newton-Raphson Method to find the root of $f(x) = x^3 - 3x + 1$.



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Reminder:

Submit your Activity # 7

<https://forms.gle/C1ti5pAMU5MF59h59>

on or before 5:30 pm.



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